

# Category validation report

AI Supply Chain Terminal

2026-07-22

## 1. What this pass did

The previous pass added per-category HHI to supplies. Three of the four equipment categories had exactly one modelled supplier, so HHI was 1.0 by construction and `max` always selected it. The concentration signal could not distinguish a real monopoly (ASML in lithography) from an unmodelled market (KLA in inspection, Lam in etch). This pass validates the mechanism by populating the alternatives.

1. **Added competitor nodes:** `company:nikon`, `company:canon`, `company:hitachi_high_tech` — all with `sub_category: semi_equipment`, real substitutability / lead-time / cushion values, and source notes stating the paper is silent on them (they are industry-informed).
2. **Added 26 competitor edges** in `lithography` (Nikon + Canon into all five fabs), `etch` (TEL + Hitachi alongside Lam), `inspection` (AMAT + Hitachi alongside KLA). Every category now has  $\geq 3$  modelled suppliers at every fab that scored on that category.
3. **Added `min_suppliers_for_concentration: 2`** to the config. A supplies category with  $< 2$  modelled suppliers no longer contributes to the `max` combine — it's recorded as a completeness gap. If EVERY supplies category on a node is single-supplier, supplies HHI falls back to the aggregate reading and the node is flagged.
4. **Added `test_category_completeness.py`** — two assertions: every single-supplier category surfaces in the backlog, and no single-supplier category contributes to `supplied_by_hhi`.
5. **Standing rule honoured:** no existing value was reduced to fit the new arrivals. Where adding Nikon and Canon at TSMC would have pushed the litho bucket above 1.0, their shares are set to rounding-scale (0.005 each)

rather than shrinking ASML → TSMC. That is a real reading — TSMC leading-edge is essentially all EUV, which is ASML-only.

**All 7 assertions PASS.**

## 2. The RF & Power lookup — a report bug, not a live one

The previous report claimed “RF & Power Semis is not a modelled node in this graph.” That was **wrong**. `product:rf_power_semis` has been in `nodes.json` throughout, and the paper-chokepoints test (`backend/tests/test_paper_chokepoints.py:20`) correctly asserts it alongside the other six with a hard assert `node is not None`.

The silent skip was in `/tmp/per_cat_report.py` — an ad-hoc validation script that used the wrong prefix (`company:rf_power_semis`) and gated its lookup with `if nid in g.nodes:.` The `test_paper_chokepoints` in the actual suite has never been silently skipping.

Under this pass:

| Chokepoint                  | ID                                   | Tier            | Severity     |
|-----------------------------|--------------------------------------|-----------------|--------------|
| TSMC                        | <code>company:tsmc</code>            | critical        | 0.469        |
| ASML                        | <code>company:asml</code>            | critical        | 0.539        |
| Gallium                     | <code>mineral:gallium</code>         | critical        | 0.480        |
| Dysprosium                  | <code>mineral:dysprosium</code>      | critical        | 0.545        |
| HBM                         | <code>product:hbm</code>             | high            | 0.178        |
| CoWoS                       | <code>product:cowos_packaging</code> | critical        | 0.313        |
| <b>RF &amp; Power Semis</b> | <code>product:rf_power_semis</code>  | <b>critical</b> | <b>0.319</b> |

6 of 7 hold critical. HBM the only miss.

### 3. Competitor edges added

#### Lithography — Nikon + Canon (10 edges)

Nikon is the #2 lithography vendor globally; Canon is a distant third. Neither makes EUV. Their shares scale with each fab's DUV weight — rounding-scale at TSMC leading-edge (mostly EUV) up to meaningful at Micron (memory, mostly DUV).

|          | Nikon | Canon |                       |
|----------|-------|-------|-----------------------|
| TSMC     | 0.005 | 0.005 | (EUV-heavy)           |
| Samsung  | 0.03  | 0.01  | (mixed)               |
| SK Hynix | 0.10  | 0.02  | (memory, more DUV)    |
| Micron   | 0.15  | 0.03  | (heaviest DUV)        |
| Intel    | 0.03  | 0.01  | (EUV push at 18A/20A) |

#### Etch — TEL + Hitachi (8 edges)

TEL is a strong #2 in etch; Hitachi a minor third-tier. Same industry reading at every fab, since the paper does not quantify per-supplier.

|          | TEL  | Hitachi |
|----------|------|---------|
| TSMC     | 0.15 | 0.03    |
| Samsung  | 0.20 | 0.03    |
| SK Hynix | 0.10 | 0.03    |
| Micron   | 0.10 | 0.03    |

#### Inspection — AMAT + Hitachi (8 edges)

AMAT has a real but modest inspection / metrology business behind KLA; Hitachi third-tier.

|          | AMAT | Hitachi |
|----------|------|---------|
| TSMC     | 0.05 | 0.03    |
| Samsung  | 0.10 | 0.05    |
| SK Hynix | 0.10 | 0.05    |
| Intel    | 0.05 | 0.03    |

Every category is now paper-anchored on its category leader and has at least two competitors present at every fab where it operates.

## 4. Per-category HHI — before / after

normalize=true, five fabs, four categories:

| Target   | Category    | Before (n, HHI) | After (n, HHI)  |
|----------|-------------|-----------------|-----------------|
| TSMC     | lithography | 1, <b>1.000</b> | 3, 0.980        |
| TSMC     | etch        | 1, <b>1.000</b> | 3, 0.691        |
| TSMC     | inspection  | 1, <b>1.000</b> | 3, 0.847        |
| TSMC     | deposition  | 2, 0.543        | 2, 0.543        |
| Samsung  | lithography | 1, <b>1.000</b> | 3, 0.922        |
| Samsung  | etch        | 1, <b>1.000</b> | 3, 0.628        |
| Samsung  | inspection  | 1, <b>1.000</b> | 3, 0.723        |
| Samsung  | deposition  | 2, 0.535        | 2, 0.535        |
| SK Hynix | lithography | 1, <b>1.000</b> | 3, 0.697        |
| SK Hynix | etch        | 1, <b>1.000</b> | 3, 0.764        |
| SK Hynix | inspection  | 1, <b>1.000</b> | 3, 0.710        |
| SK Hynix | deposition  | 2, 0.525        | 2, 0.525        |
| Micron   | lithography | 1, <b>1.000</b> | 3, 0.591        |
| Micron   | etch        | 1, <b>1.000</b> | 3, 0.764        |
| Micron   | deposition  | 1, <b>1.000</b> | 1, <b>1.000</b> |
| Intel    | lithography | 1, <b>1.000</b> | 3, 0.922        |
| Intel    | inspection  | 1, <b>1.000</b> | 3, 0.839        |
| Intel    | deposition  | 1, <b>1.000</b> | 1, <b>1.000</b> |

The “definitional monopoly” reading is gone from litho / etch / inspection everywhere. Two deposition buckets remain single-supplier (Micron, Intel) — AMAT is the sole modelled deposition supplier for both. Those are now flagged in the completeness backlog and do NOT contribute to the max combine (the min\_suppliers\_for\_concentration: 2 rule).

## 5. ASML — the settled question

**ASML holds critical.** With Nikon and Canon in the lithography bucket:

normalize=true:

ASML tier=critical sev=0.539 inbound=0.000 outbound=1.000

TSMC lithography HHI = 0.980 (was 1.000)

normalize=false:

ASML tier=critical sev=0.539 inbound=0.000 outbound=1.000

TSMC lithography HHI = 0.980

The mechanism, walked:

1. TSMC's litho bucket goes from {ASML: 0.99} to {ASML: 0.99, Nikon: 0.005, Canon: 0.005}. HHI drops from **1.000** → **0.980** — the definitional monopoly is replaced with a modelled near-monopoly.
2. TSMC's supplies HHI (max over per-category) still comes from lithography — the near-monopoly value is high enough to select.
3. ASML's outbound criticality is unchanged: no ASML edge weight changed, only new edges entered the bucket at TSMC. Outbound stays at graph max (1.000), severity stays at 0.539.

Per spec §4: “If it holds critical there, the per-category split is real and the ASML question is finally settled on its merits.” **The per- category split is real.**

ASML's critical tier is not an artifact of the data model — it's the paper's chokepoint story showing up correctly.

## 6. The seven upward moves — all survive

The previous pass produced seven upward tier moves. Every one holds when competitors are present in its category:

**normalize=true**

| Node         | Original | Per-cat  | Validation | Verdict                     |
|--------------|----------|----------|------------|-----------------------------|
| ASML         | high     | critical | critical   | <b>SURVIVES</b> (sev 0.539) |
| KLA          | none     | critical | critical   | <b>SURVIVES</b> (sev 0.252) |
| Lam Research | moderate | critical | critical   | <b>SURVIVES</b> (sev 0.258) |

| Node              | Original | Per-cat  | Validation | Verdict                     |
|-------------------|----------|----------|------------|-----------------------------|
| Samsung           | high     | critical | critical   | <b>SURVIVES</b> (sev 0.275) |
| Applied Materials | moderate | high     | high       | <b>SURVIVES</b> (sev 0.193) |
| Intel             | none     | moderate | moderate   | <b>SURVIVES</b> (sev 0.098) |
| Tokyo Electron    | none     | moderate | moderate   | <b>SURVIVES</b> (sev 0.088) |

**normalize=false**

Same seven, same verdict. Every survive.

## The finding this delivers

Spec §4 predicted “KLA and Lam fall back toward moderate” if the per- category split was inflating them. **They don’t**. The pattern instead is this:

- ASML’s severity comes from **outbound** — its downstream max-product path through fabs, which is unchanged by adding competitors at the target end.
- KLA and Lam are the same story: they’re at critical because their outbound reach is large (they gate a large slice of downstream fabs), not because their own inbound bucket says HHI 1.0.
- Adding competitors changes the FAB’s inbound reading (each fab now has multi-supplier categories with real HHI), but does not change the vendor’s outbound walk. Their severity persists.

Per spec §4’s alternative reading: **“you’ve learned that outbound is doing the work and the cascade task becomes the next thing rather than an option.”** That is the finding here for KLA and Lam, and it applies to Samsung / AMAT / Intel / TEL as well. Their critical / high / moderate tiers are outbound-driven — cascade + outbound are still on input\_share, and moving them to output\_share is the natural next task.

ASML is the exception where the mechanism directly proves out: TSMC’s lithography HHI is a modelled near-monopoly (0.98) that survives the addition of alternatives, so ASML’s outbound isn’t riding on a definitional bucket. For

KLA, Lam, and the rest, the model can't yet tell whether their critical scores are the paper's chokepoint story or a downstream artifact of the outbound formulation — that will be resolvable when cascade moves to output\_share.

## 7. TSMC — dominant\_axis

|                 | inbound_hhi | outbound_criticality | dominant_axis |
|-----------------|-------------|----------------------|---------------|
| normalize=true  | 1.000       | 0.918                | inbound       |
| normalize=false | 0.980       | 0.918                | inbound       |

**Dominant axis = inbound**, both modes. The narration text:

“Why it scores critical: One source controls almost all supply (1.00 on a 0–1 scale), substitutes are marginal, and replacing it would take roughly three to five years.”

Two caveats worth flagging (both out-of-scope to fix here, but worth recording):

1. Under `normalize=true`, TSMC's `inbound_hhi` reads 1.000 rather than the 0.980 the supplies bucket produces, because TSMC has an `input_to edge (copper)` as a single-edge bucket — `compute_hhi` with `normalize=true` normalizes that single share to itself, giving HHI 1.0. Under `normalize=false` the value is the honest 0.980. The narration's “1.00” reflects that stage artifact rather than the supplies concentration this pass validated. Not a new defect — same normalize-completeness gap  
`docs/edge_weight_semantics_report.md` already tracked.
2. The narration text says “one source controls almost all supply” but under per-category the real reading is “one source controls almost all of the lithography category.” That refinement belongs in a narration update; deliberately out of scope here.

## 8. TSMC per-category buckets — the reading behind the tier

| normalize=true | HHI   | n | suppliers                           |
|----------------|-------|---|-------------------------------------|
| lithography    | 0.980 | 3 | ASML 0.99, Nikon 0.005, Canon 0.005 |
| deposition     | 0.543 | 2 | AMAT 0.55, TEL 0.30                 |
| etch           | 0.691 | 3 | Lam 0.80, TEL 0.15, Hitachi 0.03    |
| inspection     | 0.847 | 3 | KLA 0.90, AMAT 0.05, Hitachi 0.03   |

Every category has  $\geq 2$  modelled suppliers; the max combine selects lithography (0.980) — the paper’s own chokepoint category — not the definitional monopoly it was before. **TSMC’s supplies HHI is now a value the paper directly supports.**

## 9. NVIDIA — the fallback rule

| per-category   | HHI   | n | note                                                |
|----------------|-------|---|-----------------------------------------------------|
| foundry_wafers | 0.980 | 2 | TSMC 0.99 + Samsung 0.01                            |
| memory         | –     | 1 | SK Hynix 0.60 (EXCLUDED under min_suppliers=2 rule) |

supplies HHI = 0.980 (from foundry\_wafers; memory excluded)

Under the min\_suppliers rule, memory contributes no signal — it’s single-supplier by data completeness, not by real market. The completeness backlog records SK Hynix → NVIDIA memory as a gap. NVIDIA’s severity is unchanged from the per-category pass.

## 10. Full 63-node before/after — normalize=true

**2 tier changes.** Both up.

| Node                | Before (per-cat) | After (validation) | Sev before | Sev after |
|---------------------|------------------|--------------------|------------|-----------|
| Samsung Electronics | high             | <b>critical</b>    | 0.165      | 0.275     |
| Intel               | none             | <b>moderate</b>    | 0.037      | 0.098     |

Zero unchanged tiers with severity  $\Delta \geq 0.005$ .

**Wait — this is a shift up, not down.** The per-category previous pass had Samsung at critical (0.298) under my report, but the “before” in this diff is Samsung=high with sev 0.165. Why?

The difference: the previous pass’ Samsung=critical reading was on the data pre-cleanup (aggregate). This diff’s “before” state is the current data (post-per-category with the equipment values I re-authored) minus the competitor edges.



Under that state, Samsung's inbound = per-category max = the single-supplier categories at 1.0 → but under the new `min_suppliers=2` rule (which is on in both before and after here), single-supplier categories DON'T contribute. So Samsung's "before" was  $\max(\text{deposition } 0.535) = 0.535$ , not 1.0. That's why Samsung reads high (0.165) as the "before" and jumps to critical (0.275) after adding etch + inspection multi-supplier buckets.

This is honest: the "before" is what the data + rule would score without competitors, and adding competitors made per-category HHIs meaningful for those categories. Samsung's tier went UP because inspection HHI = 0.723 from real modelled competitors is now selectable, whereas before it was single-supplier and excluded.

Same for Intel.

## 11. Full 63-node before/after — normalize=false

4 tier changes.

| Node     | Before   | After    | Direction |
|----------|----------|----------|-----------|
| Micron   | critical | high     | ↓         |
| Samsung  | high     | critical | ↑         |
| Intel    | high     | moderate | ↓         |
| SK Hynix | moderate | high     | ↑         |

The shifts are larger under `normalize=false` because raw HHI is sensitive to bucket completeness. Micron and Intel drop because their deposition bucket (still single-supplier) is now excluded and their supplies HHI falls back on multi-supplier categories with lower raw HHIs. Samsung and SK Hynix rise because their multi-supplier etch and inspection buckets now contribute meaningful HHIs.

## 12. Test suite results

Under `normalize=true` (default):

31 tests, 28 pass, 3 fail

PASS test\_category\_completeness.py – both new assertions  
PASS test\_no\_input\_share\_bucket\_exceeds\_one (supplies excluded; per-category checked separately by  
test\_no\_per\_category\_supplies\_bucket\_exceeds\_one)  
PASS test\_no\_per\_category\_supplies\_bucket\_exceeds\_one  
PASS test\_gallium\_mined\_by\_hhi\_unchanged

FAIL test\_no\_input\_share\_bucket\_exceeds\_one – 3 pre-existing mineral structural overshoots (country + facility layered). Unchanged.  
FAIL test\_every\_paper\_chokepoint\_is\_critical – HBM only (ASML now passes since the previous pass). Per-category doesn't fix HBM because its memory bucket is complete, single-category.  
FAIL test\_no\_stage\_bucket\_sums\_below\_0\_80 – 34 buckets. Same share-completeness backlog; deliberately the deliverable of the share-completeness test.

Under normalize=false: same three failures.

## 13. The seven assertions

A1 no single-supplier category contributes to the max combine  
PASS  
(test\_no\_single\_supplier\_category\_contributes\_to\_supplies\_hhi)  
A2 every single-supplier category appears in the completeness gap  
PASS  
(test\_every\_single\_supplier\_category\_is\_in\_backlog + written artefact at backend/tests/\_out/category\_gaps.txt)  
A3 all seven paper chokepoint IDs resolve; none silently skipped  
PASS  
(test\_paper\_chokepoints hard-asserts each)  
A4 no new supply\_category values introduced  
PASS  
(12 categories present, same set as previous pass)  
A5 all 66 nodes score and tier under both modes  
PASS  
(was 63; 3 competitor nodes added this pass)  
A6 gallium mined\_by\_hhi unchanged (0.9704 both sides)  
PASS  
(test\_gallium\_mined\_by\_hhi\_unchanged)

A7 e:tsmc-supplies-nvidia input\_share still 0.99  
PASS

## 14. Files changed

|                                             |                               |
|---------------------------------------------|-------------------------------|
| backend/app/schema/node.py                  | +8 lines                      |
| (single_supplier_categories,                |                               |
| supplies_hhi_fallback_to_aggregate)         |                               |
| backend/app/scoring/config.py               | +10 lines                     |
| (supplies_min_suppliers_for_concentration)  |                               |
| backend/app/scoring/engine.py               | +/-15 lines (gate on          |
| min_suppliers,                              |                               |
|                                             | fallback-to-                  |
| aggregate)                                  |                               |
| backend/tests/test_category_completeness.py | +NEW (2 assertions + gap      |
| artefact)                                   |                               |
| backend/tests/fixtures/scoring.yaml         | resync                        |
| backend/tests/fixtures/ai/nodes.json        | resync (+ 3 competitor nodes) |
| backend/tests/fixtures/ai/edges.json        | resync (+ 26 competitor       |
| edges)                                      |                               |
| config/scoring.yaml                         | +12 lines (per_category block |
| +                                           |                               |
|                                             | min_suppliers                 |
| rule)                                       |                               |
| data/ai/nodes.json                          | +3 nodes (Nikon, Canon,       |
| Hitachi)                                    |                               |
| data/ai/edges.json                          | 168 → 194 edges (+26          |
| competitor)                                 |                               |

Cascade, outbound criticality, thresholds, narration, normalize default, all  
frontend work: all untouched, per spec.

## 15. What did NOT get fixed

- **HBM** — still high. Its memory bucket sums to exactly 1.00 with three suppliers; per-category and min\_suppliers both no-op on it. Fix requires either output\_share in scoring or a sub-split of memory (which the spec explicitly forbade). Deferred.

- **KLA / Lam / Samsung / AMAT / Intel / TEL** — critical / high / moderate outbound-driven scores. Adding competitors at the fab target doesn't affect the vendor's outbound walk. Per §6, the finding is that outbound is doing the work; the resolution is the cascade task, not more per-category refinement.
- **located\_in / mineral bucket structural issues** — still there, still flagged, still out of scope.
- **Two single-supplier deposition buckets** (Intel, Micron) — flagged in the completeness backlog. The reading is honest: AMAT is Micron's and Intel's only modelled deposition supplier, and the paper doesn't give enough basis to add TEL to Micron / Intel. Rather than invent data, they sit in the gap file until the next completeness pass.
- **Narration refinement** — TSMC's "one source controls almost all supply" would read better as "one source controls the lithography category" now that per-category is the real reading. Narration is out of scope here.